PROBLEM VF3

Annular Flow with a Moving Outer Cylinder

Original statement · VF3.pdf

Annulus | Poiseuille + Couette | Zero flux

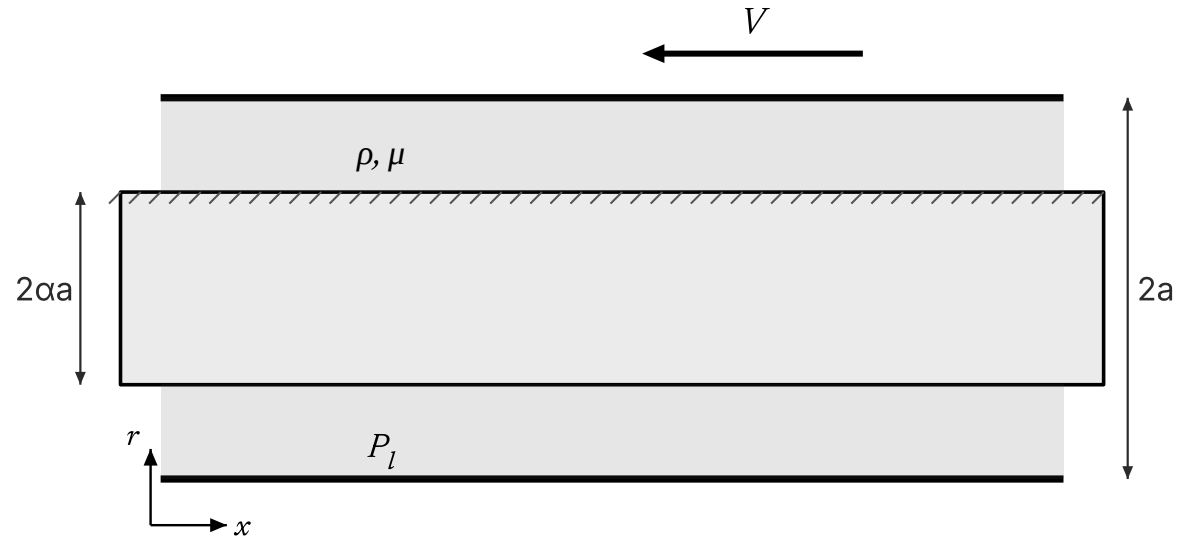
OFFICIAL PROBLEM STATEMENT

A fluid of density ρ and viscosity μ is confined between two infinitely long coaxial cylinders of radii a and αa, with $\alpha < 1$, aligned in the x direction. The fluid is forced to move steadily due to the presence of a known reduced pressure gradient $P_l = -\partial P/\partial x$ and also due to the backward motion of the outer cylinder, with velocity V.

1. Determine the fluid velocity $v_x(r)$.

2. Obtain the power needed to move the outer cylinder.

3. Calculate the value of V for which the volume flux in the pipe is identically zero.



Statement figure (redrawn from VF3.pdf): liquid of density ρ and viscosity μ between a rod of radius αa and a cylinder of radius a, along x, driven by the pressure gradient P_l; the outer cylinder moves backward with velocity V

Source: VF3.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 7: unidirectional flow, Eq. 7.7; steady flow in a circular pipe, Eqs. 7.13-7.16.

Worked solution

Hypotheses, Data and Unknowns

- Steady, fully developed flow of a liquid with constant μ: $\bar{v} = v_x(r)\bar{e}_x$ (Notes.pdf, Eq. 7.1, with axial symmetry); P_l is a known constant (Notes.pdf, p. 74).
- The inner body, a rod of radius αa, is at rest; the outer cylinder of radius a moves backward, that is, with velocity $-V\bar{e}_x$.
- "Backward" means against the pressure-driven flow: $P_l > 0$ drives the liquid towards $+x$.
- The power is taken per unit length of cylinder, since the cylinders are infinitely long.

Symbol	Meaning	SI unit
a, αa	outer and inner radius ($\alpha < 1$)	m
P_l	reduced pressure gradient, $-\partial P/\partial x$	Pa/m
V	speed of the outer cylinder (towards $-x$)	m/s
μ	viscosity	kg/(m s)

Velocity Profile

Equation (7.7) in steady state, with the conditions at the two walls:

$$0 = P_l + \frac{\mu}{r}\frac{d}{dr}\left(r\frac{dv_x}{dr}\right); \qquad r = \alpha a:\ v_x = 0, \qquad r = a:\ v_x = -V$$

Multiplying by r/μ and integrating once: $r\,dv_x/dr = -P_l r^2/(2\mu) + C_1$. Dividing by r and integrating again:

$$v_x = -\frac{P_l}{4\mu}r^2 + C_1 \ln r + C_2$$

The term $C_1 \ln r$ cannot be dropped: the axis is not part of the domain, so there is no regularity condition. It is convenient to write the solution with the inner wall as origin of the logarithm, $v_x = -\frac{P_l}{4\mu}\left(r^2 - \alpha^2 a^2\right) + C \ln \frac{r}{\alpha a}$, which satisfies $v_x(\alpha a) = 0$ for any C. The outer wall fixes C:

$$-V = -\frac{P_l}{4\mu} a^2 \left(1 - \alpha^2\right) + C \ln \frac{1}{\alpha} \implies C = \frac{-V + \dfrac{P_l}{4\mu} a^2 \left(1 - \alpha^2\right)}{\ln\left(\alpha^{-1}\right)}$$

$$v_x = -\frac{P_l}{4\mu}\left[r^2 - (\alpha a)^2\right] + \left[-V + \frac{P_l}{\mu}\frac{\left(1 - \alpha^2\right) a^2}{4}\right] \frac{\ln\left(r/\alpha a\right)}{\ln\left(\alpha^{-1}\right)}$$

Check: at $r = \alpha a$ the logarithm is zero and the bracket $r^2 - (\alpha a)^2$ vanishes; at $r = a$ the first term is $-P_l a^2(1 - \alpha^2)/(4\mu)$ and the second is $-V + P_l a^2(1 - \alpha^2)/(4\mu)$, whose sum is $-V$.

Power Needed to Move the Outer Cylinder

The force per unit area of the liquid on the outer wall is $\tau_p = \bar{\bar{\tau}}' \cdot \bar{n}$ with $\bar{n}$ pointing from the wall into the liquid, $\bar{n} = -\bar{e}_r$, so $\tau_p = -\mu \left(\partial v_x / \partial r\right)_{r=a}$ (the convention of (7.16)). Differentiating,

$$\frac{dv_x}{dr} = -\frac{P_l}{2\mu} r + \frac{C}{r} \implies \tau_p = \frac{P_l a}{2} - \frac{\mu C}{a} = \frac{P_l a}{2} - \frac{\mu}{a}\frac{-V + \dfrac{P_l}{4\mu} a^2 \left(1 - \alpha^2\right)}{\ln\left(\alpha^{-1}\right)}$$

$$\tau_p = \frac{P_l a}{2}\left[1 - \frac{1 - \alpha^2}{2 \ln\left(\alpha^{-1}\right)}\right] + \frac{\mu V}{a \ln\left(\alpha^{-1}\right)}$$

The liquid pushes the cylinder with the force $2\pi a\,\tau_p$ per unit length, in the direction $+x$. The external agent that moves the cylinder at velocity $-V\bar{e}_x$ must balance it, with the force $-2\pi a\,\tau_p\,\bar{e}_x$, so the power it supplies per unit length is force times velocity:

$$W = (-2\pi a\tau_p)\,(-V) = \tau_p\,V\,2\pi a$$

(power per unit length of cylinder, in W/m).

Zero Volume Flux

The flux through the annulus is $Q = \int_{\alpha a}^{a} 2\pi r\, v_x\, dr$. With $G = P_l/(4\mu)$, $b = \alpha a$, $\mathcal{L} = \ln(\alpha^{-1})$ and the two integrals (Barrow's rule):

$$\int_b^a r\left(r^2 - b^2\right) dr = \left[\frac{r^4}{4} - \frac{b^2 r^2}{2}\right]_b^a = \frac{\left(a^2 - b^2\right)^2}{4}$$

$$\int_b^a r \ln\frac{r}{b}\, dr = \left[\frac{r^2}{2}\ln\frac{r}{b} - \frac{r^2}{4}\right]_b^a = \frac{a^2}{2}\mathcal{L} - \frac{a^2 - b^2}{4}$$

(the second uses $\int r\ln r\, dr = \frac{r^2}{2}\ln r - \frac{r^2}{4}$, checked by differentiating). With $D_a = a^2 - b^2 = a^2(1-\alpha^2)$ and $C = (-V + GD_a)/\mathcal{L}$:

$$\frac{Q}{2\pi} = -G\frac{D_a^2}{4} + C\left(\frac{a^2}{2}\mathcal{L} - \frac{D_a}{4}\right) = 0$$

The key stops at the equivalent integral equation, boxed, $\frac{V}{\ln(\alpha^{-1})}\int_{\alpha a}^{a} r\ln\frac{r}{\alpha a}\, dr = \frac{P_l}{4\mu}\int_{\alpha a}^{a}\left[\frac{(1-\alpha^2)a^2\ln(r/\alpha a)}{\ln(\alpha^{-1})} - r^2 + (\alpha a)^2\right] r\, dr$, and does not evaluate it. Evaluating it (computed here) and solving for V:

$$(-V+GD_a)\left(\frac{a^2}{2}-\frac{D_a}{4\mathcal{L}}\right)=G\frac{D_a^2}{4}\implies V=GD_a\,\frac{\dfrac{a^2}{2}-\dfrac{D_a}{4}-\dfrac{D_a}{4\mathcal{L}}}{\dfrac{a^2}{2}-\dfrac{D_a}{4\mathcal{L}}}$$

and, using $\frac{a^2}{2}-\frac{D_a}{4}=\frac{a^2(1+\alpha^2)}{4}$ and multiplying numerator and denominator by $4\mathcal{L}/a^2$:

$$V=\frac{P_l a^2}{4\mu}\left(1-\alpha^2\right)\frac{\left(1+\alpha^2\right)\ln\left(\alpha^{-1}\right)-\left(1-\alpha^2\right)}{2\ln\left(\alpha^{-1}\right)-\left(1-\alpha^2\right)}$$

Answers and Checks

- Velocity: $v_x=-\frac{P_l}{4\mu}\left[r^2-(\alpha a)^2\right]+\left[-V+\frac{P_l}{4\mu}(1-\alpha^2)a^2\right]\ln(r/\alpha a)/\ln(\alpha^{-1})$.
- Power per unit length: $W=2\pi a\,V\left\{\frac{P_l a}{2}\left[1-\frac{1-\alpha^2}{2\ln(\alpha^{-1})}\right]+\frac{\mu V}{a\ln(\alpha^{-1})}\right\}$.
- Zero flux: $V=\frac{P_l a^2}{4\mu}(1-\alpha^2)\frac{(1+\alpha^2)\ln(\alpha^{-1})-(1-\alpha^2)}{2\ln(\alpha^{-1})-(1-\alpha^2)}$.
- Limit $\alpha\to 0$ (no rod): $\ln(\alpha^{-1})\to\infty$ and $V\to\frac{P_l a^2}{4\mu}\cdot\frac{1}{2}=\frac{P_l a^2}{8\mu}$; directly, a pipe with a wall moving at $-V$ has $Q=\pi a^4 P_l/(8\mu)-V\pi a^2=0$, the same V.
- Limit $\alpha\to 1$ (thin gap $h=a(1-\alpha)$): with $\epsilon=1-\alpha$, $\ln(\alpha^{-1})\simeq\epsilon+\epsilon^2/2+\epsilon^3/3$, the numerator of the last fraction is $\simeq 2\epsilon^3/3$ and its denominator $\simeq 2\epsilon^2$, so $V\to\frac{P_l a^2}{4\mu}\,2\epsilon\,\frac{\epsilon}{3}=\frac{P_l h^2}{6\mu}$, which is the plane result: $Q=P_l h^3/(12\mu)-Vh/2=0$.
- Dimensions: $P_l a^2/\mu$ is $ML^{-2}T^{-2}\cdot L^2/(ML^{-1}T^{-1})=LT^{-1}$, a velocity; $\tau_p V$ per unit length is $(\mathrm{N/m^2})(\mathrm{m/s})(\mathrm{m})=\mathrm{W/m}$.
- Check against the official key: VF3.pdf gives the same velocity profile, the same τ_p and $W=\tau_p V\,2\pi a$, and, for the zero flux, the integral condition (not evaluated).